

Analysis of Existing Plastic Roads and Suitability of Dry or Wet Mix Technology for Different Types of Plastic Waste

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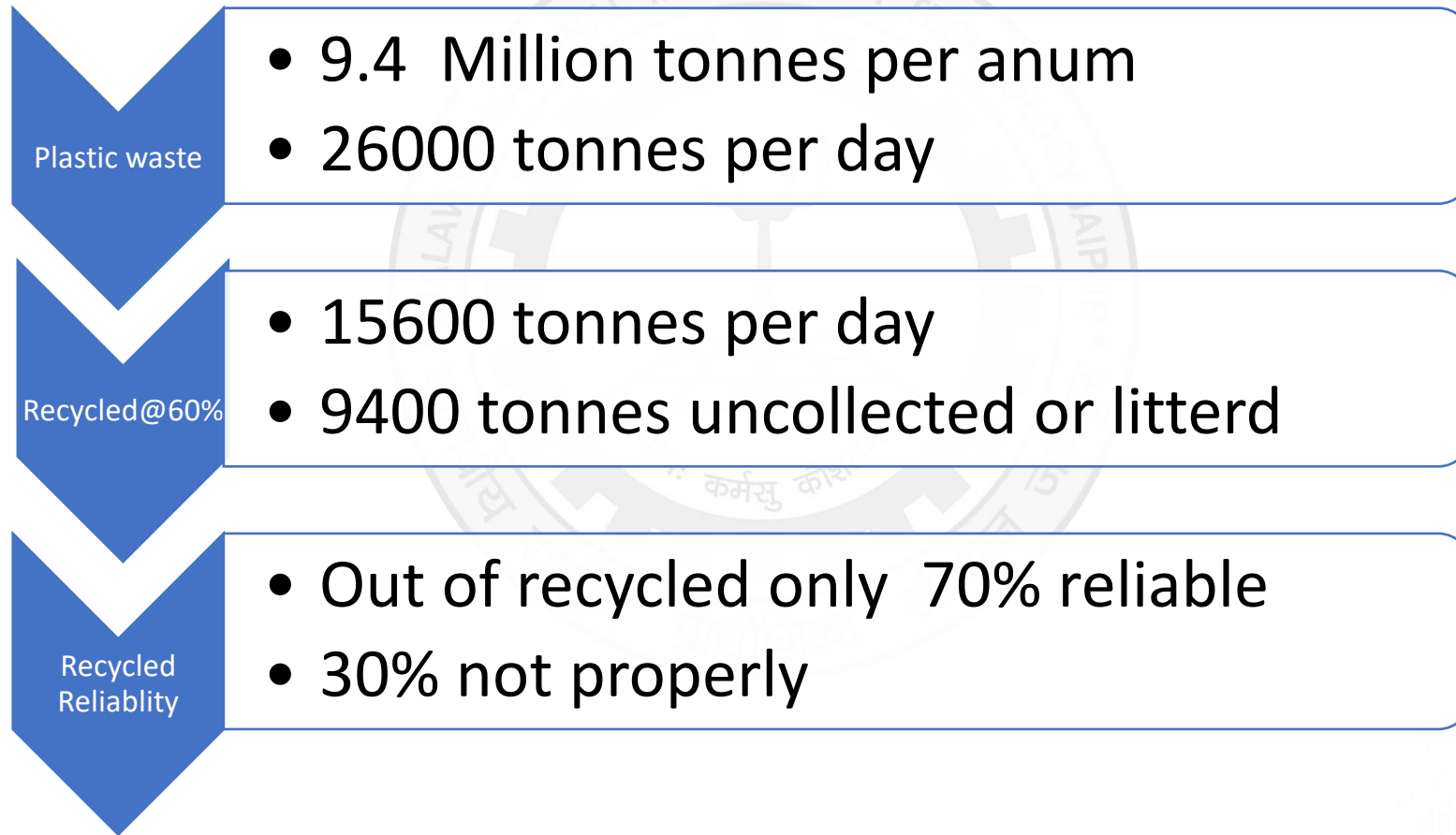
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Plastic waste : Stats in India



Type of Plastics

The Society of the Plastics Industry, Inc. (SPI) introduced its resin identification coding system in 1988 at the urging of recyclers around the country.

The seven types of plastic include:

1. Polyethylene Terephthalate (PETE or PET)
2. High-Density Polyethylene (HDPE)
3. Polyvinyl Chloride (PVC)
4. Low-Density Polyethylene (LDPE)
5. Polypropylene (PP)
6. Polystyrene or Styrofoam (PS)
7. Miscellaneous plastics (includes: polycarbonate, polylactide, acrylic, acrylonitrile butadiene, styrene, fiberglass, and nylon)

A Summary of Waste Plastics that Can Be Recycled According to ASTM D7611

Recycling Symbol	Major Products		Major Physical Property and Biodegradability
Polyethylene terephthalate		Single-use drink bottles, peanut butter containers, salad dressing and vegetable oil containers	• Thermoplastic polymer resin of the polyester family, and semi-crystalline. • Density = 1.15 ± 0.03 g/cm³, tensile strength = 0.8 ± 0.14 N/mm² (Biber et al. 2019) • Biodegradable
High-density polyethylene		Milk jugs, juice bottles, bleach, detergent, and cleaner bottles, shampoo bottles, motor oil bottles	• Molecules can fit closer together, resulting in stronger, denser, and more rigid than LDPE. • High density ($0.940 - 0.965$ g/cm³); Melting point = 131°C, melt flow rate = 6.12 g/10min (Casey et al. 2008)
Polyvinyl chloride		Plumbing pipes and fitting, wire jacketing, blister packaging, cooking oil bottles, teething rings	• Contains Cl- in PVC (Lu et al. 2019) • Relatively impervious to sunlight and weather, • PVC, after many years, will result in granulation. It will never totally biodegrade.

A Summary of Waste Plastics that Can Be Recycled According to ASTM D7611

Recycling Symbol	Major Products		Major Physical Property and Biodegradability
Low-density polyethylene		sheeting, squeezable bottles, shopping bags, furniture, heavy duty sacks, general packaging	Low density ($<0.930 \text{ g/cm}^3$), softening point = 85°C, density = 0.92 g/cm^3, tensile strength = 8.96 MPa (Zhang and Hu 2016); melting index of 0.8 g/min at 190°C (Yan et al. 2015)
Polypropylene		Straws, wrapping, caps, syrup and medicine bottles, potato chip bags, packing tape and rope	Density = 0.9 g/cm^3, melt flow index = 12 g/min; mechanical properties of recycled PP depend not only to its melt flow index but also to the contaminants contained in the recycled PP from waste origin and additives added by suppliers. (Jmal et al. 2018)
Polystyrene		Disposable plates and cups, meat trays, egg cartons, carry-out containers, compact disc cases	- Density = $1.1 \pm 0.19 \text{ g/cm}^3$, tensile strength = $3 \pm 1.13 \text{ N/mm}^2$ (Biber et al. 2019) - Easily attacked by many organic solvents (e.g. acetone). Melting point: $210 - 249^\circ\text{C}$ (Rogers 2015) - Non-biodegradable.

A Summary of Waste Plastics that Can Be Recycled According to ASTM D7611

Recycling Symbol	Major Products		Major Physical Property and Biodegradability
 Plastic Other than 1 to 6		Most common e-plastics, used for electronic devices, e.g. laptops and mobile phones, pipes, children's toys .	• Thermoplastic polymer (Keenan, 2018); $(C_8H_8)_x$, terpolymer made by polymerizing styrene and acrylonitrile in the presence of polybutadiene; Very durable, stiff, and strong • Glass transition temperature = $105^{\circ}C$; no true melting point due to amorphous
		Footwear industry, including soles and insoles (Lopes et al. 2015)	• Linear molecular structure, specific gravity = 0.92, processing temperature = $65-80^{\circ}C$ (Sengoz 2008), melting flow index = 2.5 g/10 min (Mansourian et al. 2019) • Provides the modification of asphalt via a crystallization of rigid 3-D networks resulting in a considerable change in physical, chemical, and morphological properties of asphalt
		Used in the food cold chain, in upholstered furniture and mattresses, shoes, cars, medical device as well as for the thermal insulation of buildings .	• A thermostable polymer, no degradation at high temperature • Only small percentage (10%) is recycled or incinerated • Only a small amount (10%) is recycled, a little research on the use of PU wastes in asphalt pavement (Cong et al. 2019)

A Summary of Waste Plastics that Can Be Recycled According to ASTM D7611

Recycling Symbol	Major Products		Major Physical Property and Biodegradability
Plastic other than 1 to 6		CDs and DVDs	PC is one of the most important engineering plastics, possessing excellent mechanical performance, biodegradability, dimensional stability, flame resistance and high stability to different environmental conditions (Tarantili et al. 2010) • Specific gravity = 1.2, water absorption = 0.15%, compressive strength = 86.1 MPa, tensile modulus = 2.37 MPa, elongation = 110% (De la Colina Martínez et al. 2019); Melt flow index = 2.6 g/10 min, tensile strength = 69 MPa, Modulus of elasticity = 1960 MPa (Tarantili et al. 2010) • The strength of recycled PC from CDs and DVDs are only 20% lower than those of virgin PC. (Pedreno-Rojas et al. 2019); however, PC recycled from CDs are found to close to pure PC and the property is not damaged (Zhu et al. 2017)

Plastic waste in Pavements

- Asphalt is a hydrocarbon material that has similar Chemical function as plastic.
- Highway sector has a great potential to use waste in large quantity.
- Density of Most of the Plastics are in range of Bitumen.
- Waste plastics, after undergoing the process of collection, drying, sorting and shredding can be incorporated in the Bituminous mix either using a wet or dry process.

Mixing of Plastics

- **Wet process of plastic waste mixing**
- Shredded plastics are pre mixed with the bitumen before mixing it with the mineral aggregates.
- No code developed by IRC
- **Dry process of plastic waste mixing**
- Plastic waste with size between 2.36mm to 600 microns is fed with heated aggregates and then bitumen added.
- IRC-SP 98-2013 is the code for dry mixing.

Literature review : PET

Mixture Type	PET Size	Dosage	Major Findings on PET modified asphalt mixtures	Country, Reference
HMA, 60/70-pen	2.36-4.75 mm Waste PET	20, 30, 40, 50, and 60% of aggregate by volume	● 20% replacement by weight (equivalent to 5% by weight of the asphalt mix) results in a 2.8% reduction in unit weight of the mixes, and results in higher Marshall quotient, and the lowest reduction of stability.	Iran, Hassani et al. (2005)
SMA, 80/100-pen	Crushed PET passing 1.18 mm sieve	0, 2, 4, 6, 8, 10% of asphalt by weight	● Higher resilient modulus value, higher rutting resistance based on wheel tracking test, 4% PET shows lowest rut depth; Significantly reducing drain down ● Decreased tensile strength ratio (TSR), but all above 70% ● 4 to 6% waste PET of asphalt would be optimum; overall satisfying the standard requirement	Malaysia, Ahmadinia et al. (2012)
Dense graded HMA, PG 64-22	30-mm length, 0.7-1 mm width, and 0.5 -0.7 mm thickness	0.2, 0.4, 0.6, 0.8, and 1.0% of mix by weight	● PET fiber-reinforced HMA had higher air void contents (4.92-6.3%) than the plain HMA (4.82%) ● The pre-peak fracture energy increases with increasing PET fiber content up to the optimum fiber content of 0.4% ● Post-peak toughness indices of fiber-reinforced HMA is 1.3 times those of plain HMA.	South Korea, Yoo and Al-Qadi (2014)
SMA, 80/100-pen	Crushed PET particles < 2.36 mm	0, 0.5 and 1% of aggregate by weight	● The variation of PET was more influential on the stiffness of mixtures compared to the stress levels ● 0.41% of PET was selected as the optimal PET value to reach the highest stiffness for the SMA mixtures ● Improved permanent deformation resistance	Malaysia, Moghaddam et al. (2014, 2015)

HMA, 60/70- pen	Waste PET	10, 20, and 30% of aggregate by weight	● Reduced density, increased air void, VMA, and Marshall stability	Nigeria, Sojobi, et al. (2016)
HMA, 60/70- pen	Waste PET	0, 1.68, 2.52, 3.36, and 5.04% of mix by weight	● Marginal change in bulk specific gravity: 0.2% decrease compared to unmodified mixture ● Marshall quotient increased by 21-22%	Trinidad and Tobago, Maharaj et al. (2018)
RAP binder	PET derived additive through aminolysis	2% of asphalt by weight	● Improved anti-stripping property, increased rutting resistance by at least 15% and fatigue cracking resistance by up to 60% based on DSR tests ● Enhanced low temperature performance	Hong Kong China, Leng et al. (2018)
Semi dense HMA 50/70- pen	10-5 mm for coarse fraction, 2- 0.25 mm for fine fraction	10 and 20% fine fraction	● PET modified HMA showed higher energy dissipation, resilient modulus, and better rutting resistance, but more susceptibility to cracking at low temperature. ● Mixtures with PET has lower affinity with aggregate, resulting in more water damage by TRS <80%. The ratio of PET scrap to binder replaced should not exceed 1:1.	Spain, Movilla Quesada et al. (2019)

Review for HDPE/LDPE

AC-20	HDPE, 4-8% by weight of asphalt	● HDPE passing 2-mm retained on 0.42-mm, mixing at 144-155 °C for 30 min ● HDPE-modified asphalt mixture increased Marshall quotient by 50%, better rutting resistance	Turkey, Hinishoglu (2004)
SMA, 40/60-pen	HDPE, 2 to 5% by weight of asphalt	● Melting point = 131°C, melt-flow rate 6.12 g/10min ● Wheel track test: recycled polymer binder has more rutting than polymer modified binder, but meets the requirement. ● Indirect tensile fatigue: polymer modified binder is better than recycled polymer (RP) and polymer fiber (PF) binder, due to a slight lack of elasticity on the part of RP binder. RP binder is better than PF. ● 4% HDPE is optimized and most promising.	Ireland, Casey et al. (2008)
HMA, 80/100-pen	HDPE from carry bags, 2.5, 5.0, 7.5% by weight of asphalt, 2 mm by 2 mm size	● String at 170°C with 3500 rpm for 25 min ● Melting point of 130°C ● Increased viscosity, lower temperature susceptibility ● More resistance to moisture based on higher TSR and improved stripping characteristics, higher stiffness by 28%, ● Significantly higher fatigue life based on dissipated energy, phase angle, and stiffness ● Field fatigue lives is 2.5 time longer than that of conventional mixture.	U.S. South Carolina, Punith et al. (2013)

35/50	HDPE, 5% by weight of asphalt, powder and granulated	● Stirring at 180°C for 60 min at 350 rpm ● Increased softening point, decreased penetration, increased viscosity	Portugal, Costa et al. (2013)
Dense graded HMA, 35/50	HDPE, 4-mm, 2, 4, 6, 8% by weight of asphalt	● Melting point = 140 -150°C ● Increases bulk density, increased compressive strength, improved moisture resistance, and improved rutting resistance based on higher Marshall quotient value	Algeria, Melbouci et al. (2014)
Glassasphalt, PG 64-16	HDPE, 0, 2, 4, 5, 8, 10% by weight of asphalt	● HDPE is converted into much smaller pieces through cryogenic process, remaining on #100 and #200 sieves. ● Mixing at 180°C and 4500 rpm within 40 min. ● Optimum dosage of HDPE is about 6-8% based on fatigue, creep, and resilience tests. ● HDPE is depolymerized in reaction to asphalt, and it absorbs the aromatic part of asphalt and swallows.	Iran, Arabani and Pedram (2016)
35/50 pen	HDPE, 5% by weight of asphalt	● 4-mm maximum, reduce to a powder passing a 0.45 mm sieve mesh by Frozen with liquid nitrogen and then milled ● Mixing at 600 rpm for 1 hour at 180°C	Portugal, Costa et al. (2019)

Research Gaps

- The Analysis of Constructed Plastic Roads in Actual Situation is still to be done.
- Almost all the researchers have analyzed single type of plastics in their study which generates a lacuna in comparative analysis of different plastics.
- Scope in Percentage of different plastics can be assessed by Dry and Wet Mix methods.
- Advance Rheological tests on Bitumen as well as Microscopic and Micro structural Analysis of Plastic modified bitumen.

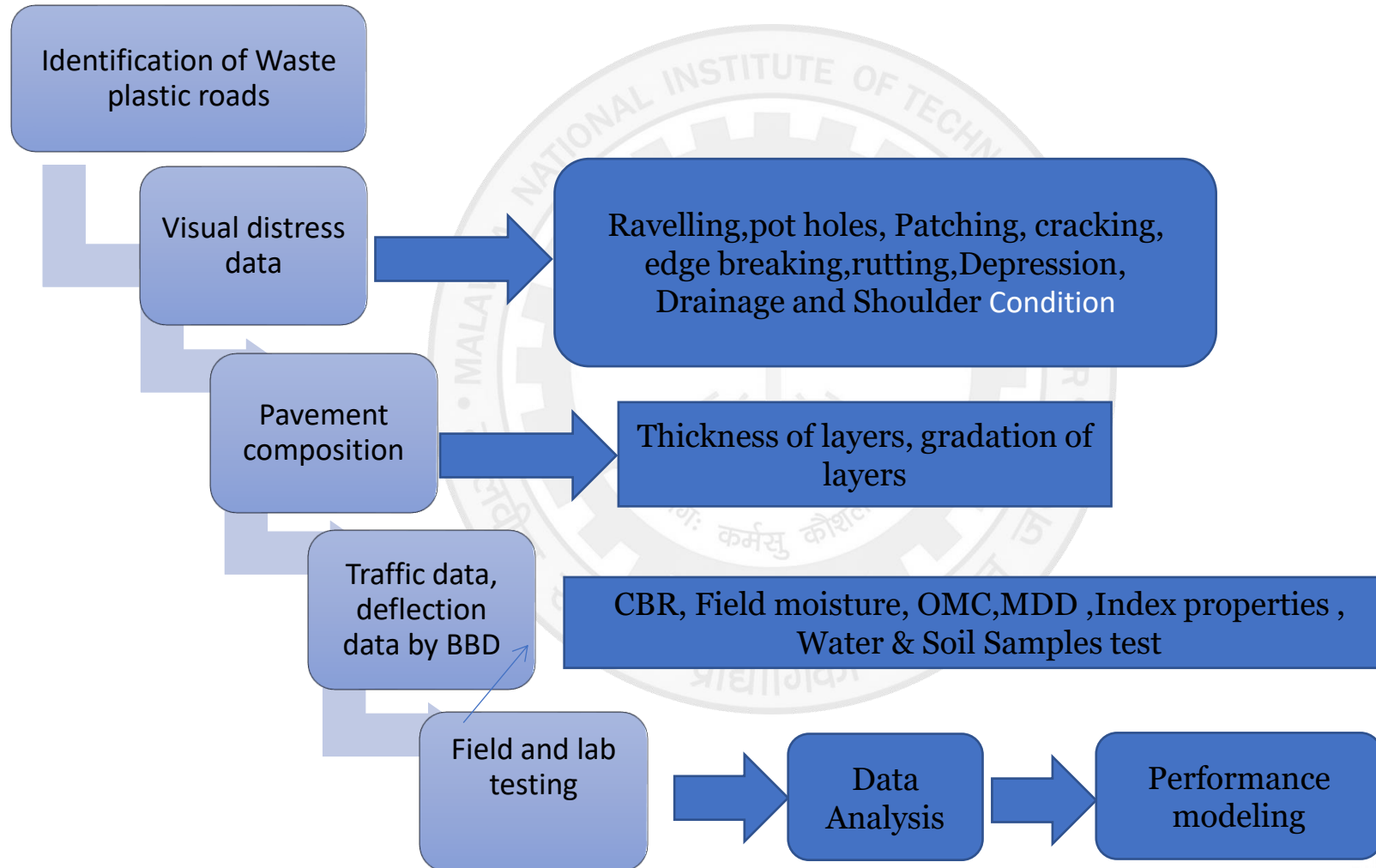
Research Gaps

- Possibility of Increased percentage of waste plastic by choosing suitable method of mixing.
- Environmental and Economical consideration for plastic waste roads for Existing ways of mixing .
- Scope of Wet and Dry mix Technologies in Rural roads.

Objectives

- Performance assessment of Existing Plastic roads by intense field investigation.
- To Analyze the Plastic Roads in comparison of Conventional Roads.
- Suitability of appropriate method of mixing for different categories of waste plastics.
- Optimum percentage of Waste plastics for both the Mixing Methods.
- Feasibility study of both the mixing methods in field application.

Research methodology : Waste plastic Roads

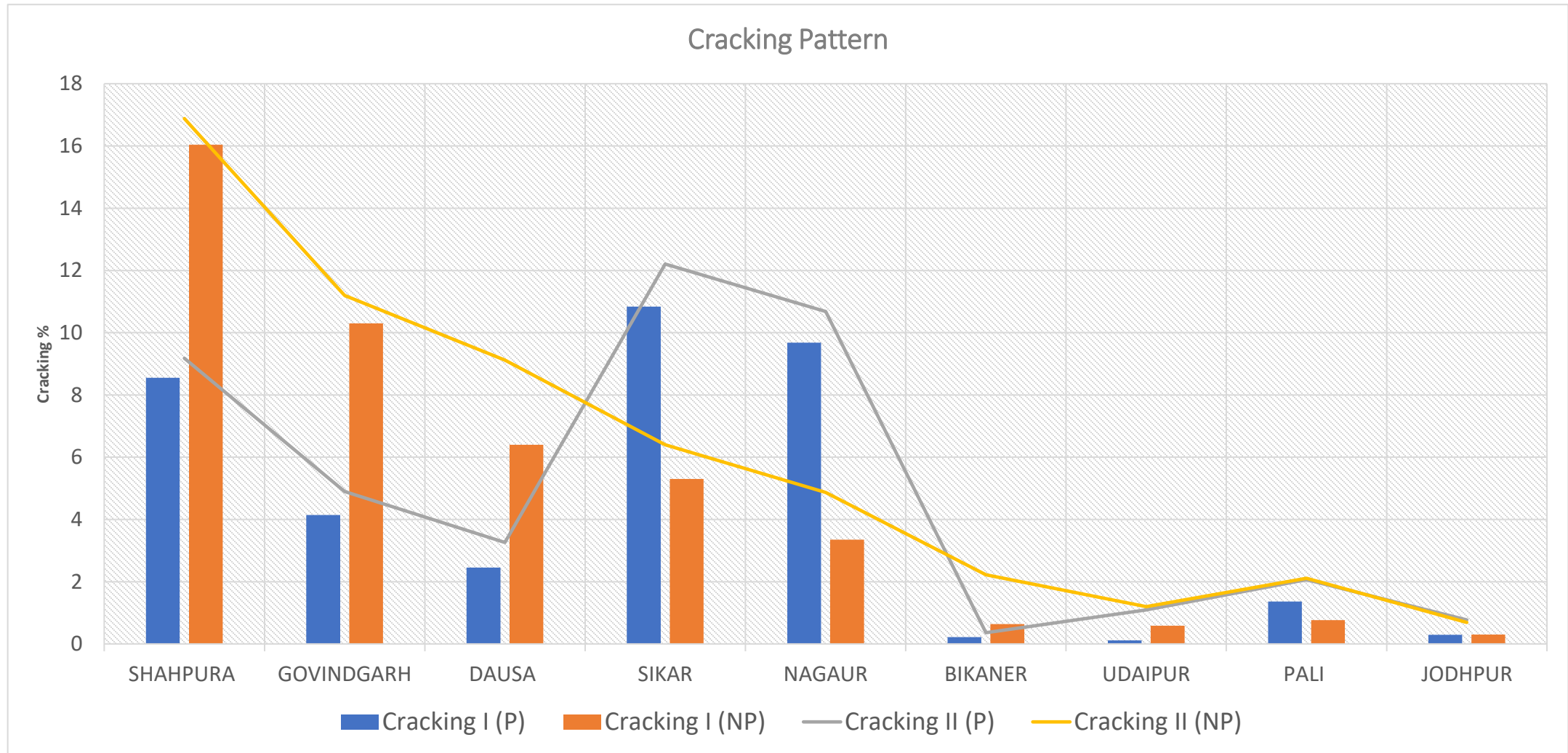


Work Progress

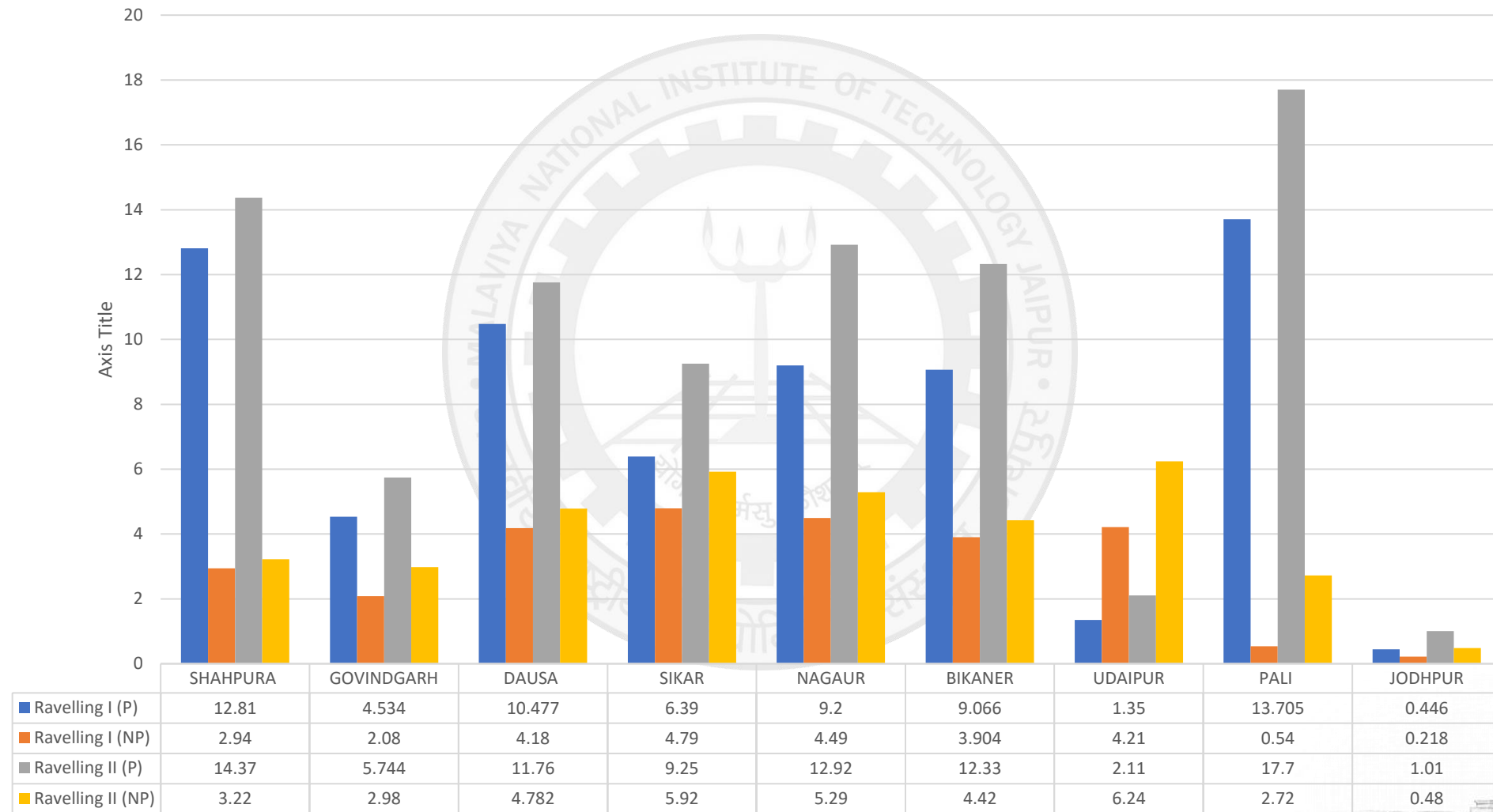
Finalization of Roads For Assessment.

S. No	District name	Block Name	Road Name	Road Sr. No.	Tech. length	Assesment length (WP)	Assesment length (conv)
1	Jaipur	Shahpura	Radawas to Mathura wali dhani	171	3.9	3	1
2	Jaipur	Govindgarh	Khejroli to Dhani Leshwa	163	4	4	1
3	Dausa	Sikrai	Gumanpura to Bairwanda	108	4.8	4	1
4	Sikar	Danta ramgarh	Khoor to Karoon ki dhani	362	6.6	5	1
5	Nagaur	Makrana	Bhaiy kalla to Gujaron ki dhani	269	4.3	4	1
6	Bikaner	Nokha	Panchoo to Garudon ki dhani	81	5.9	5	1.5
7	Udaipur	Kherwara	Larathi to Larathi B	424	6	5	1.5
8	Jodhpur	Baori	Dewaton to Raikon ki dhani	222	6	5	1
9	Pali	Bali	Thandeberi to Aamriya	283	5.5	5	1.5

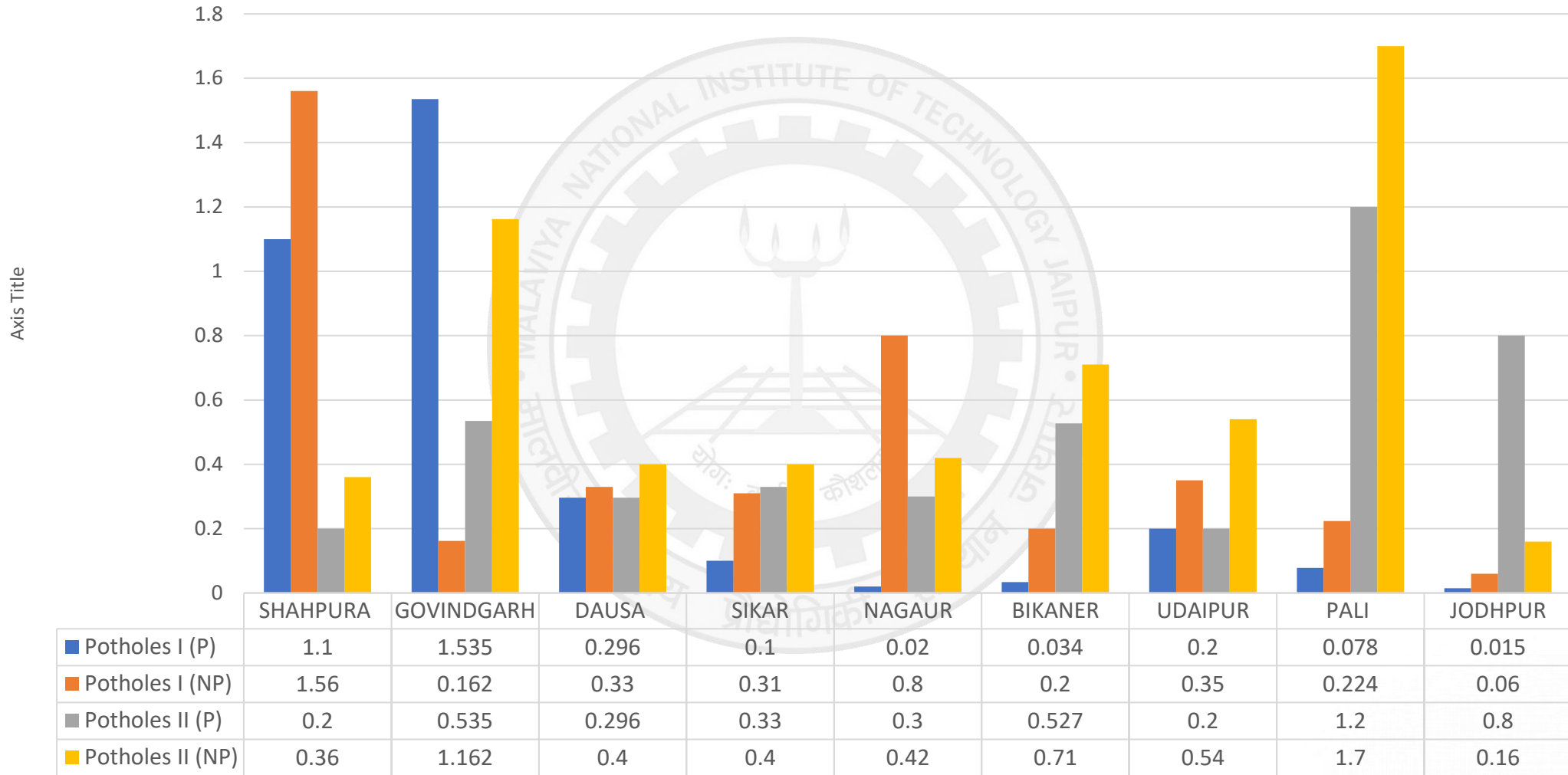
Visual Distress Data



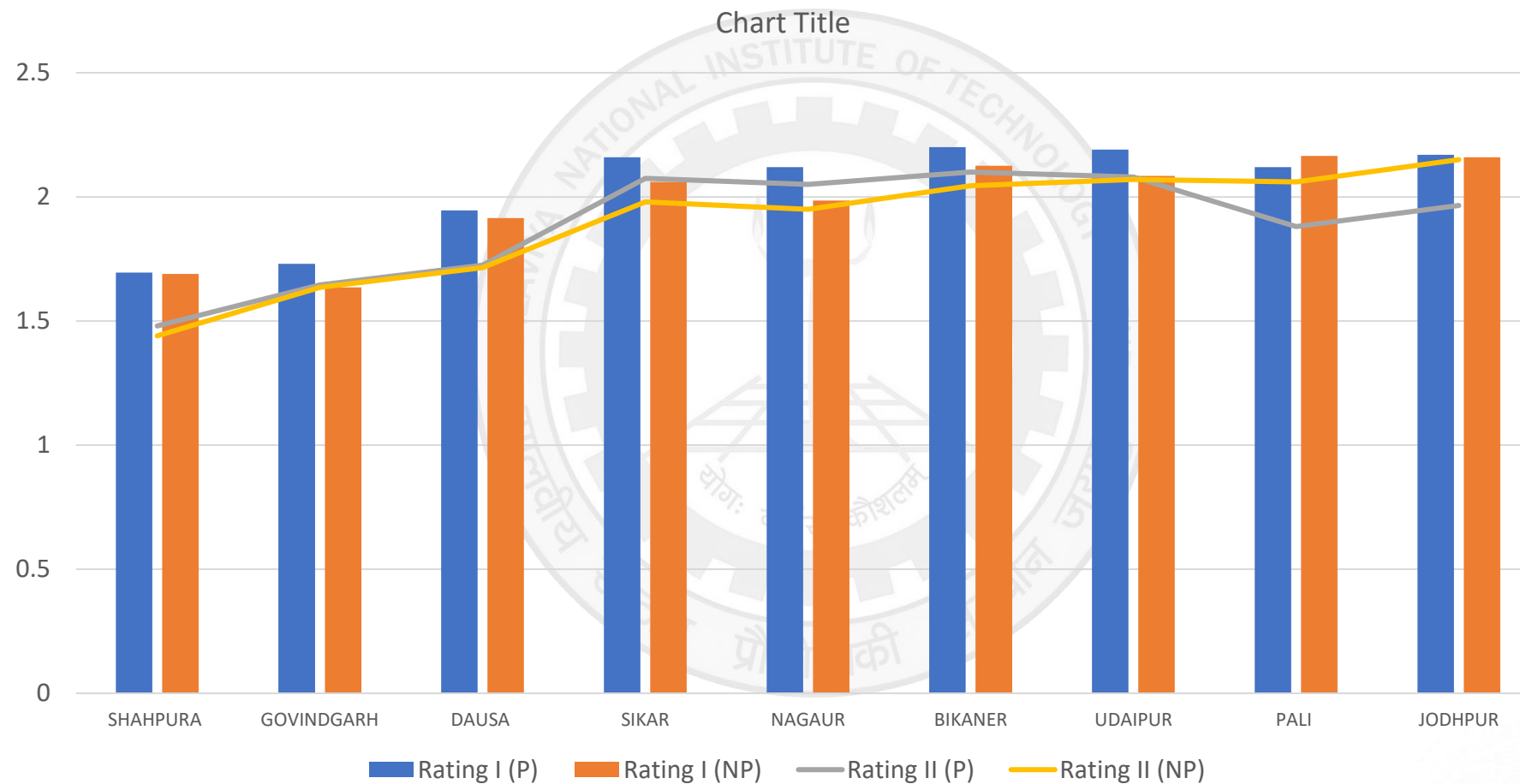
Ravelling Pattern



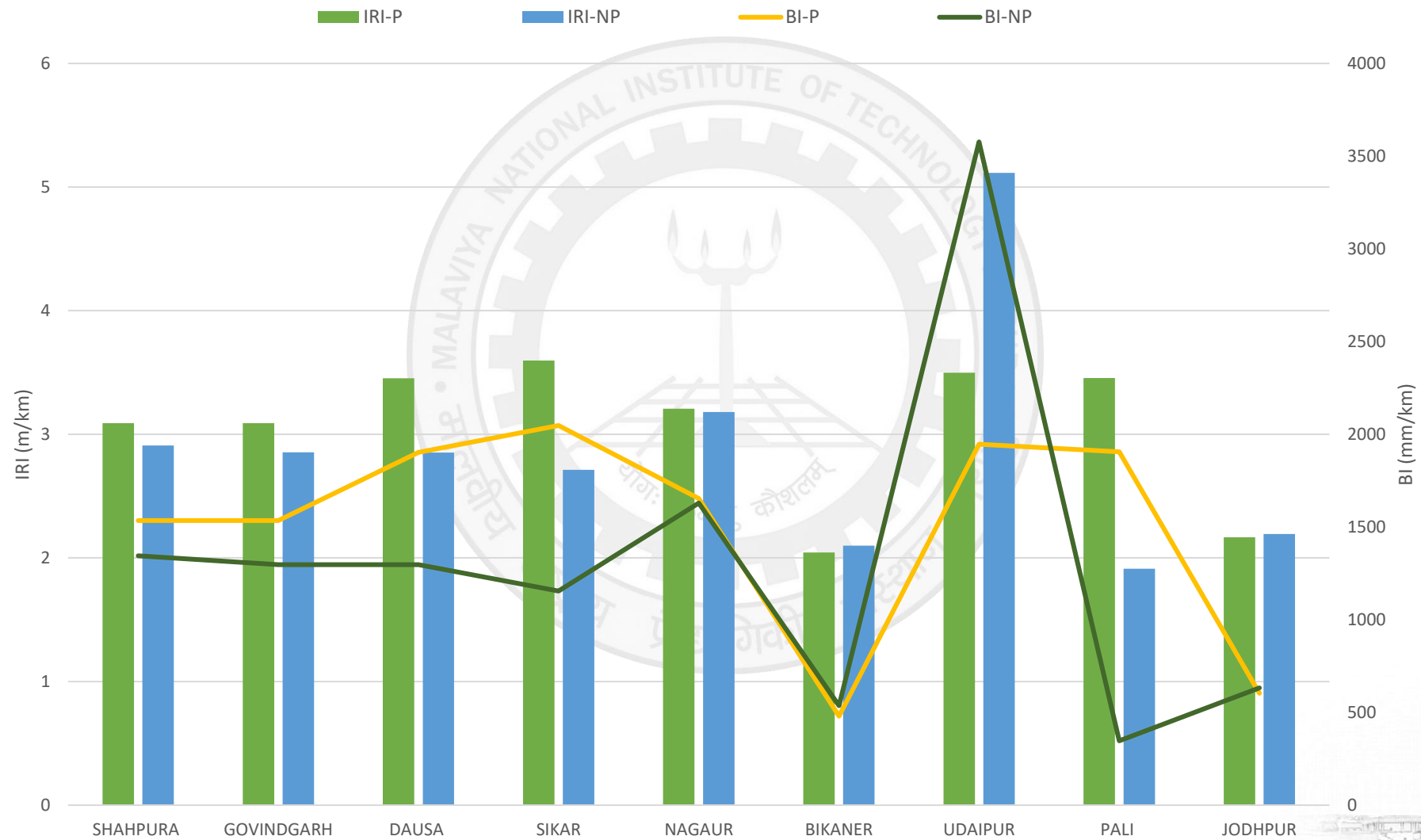
Pot Holes Comparison



PCI ratings



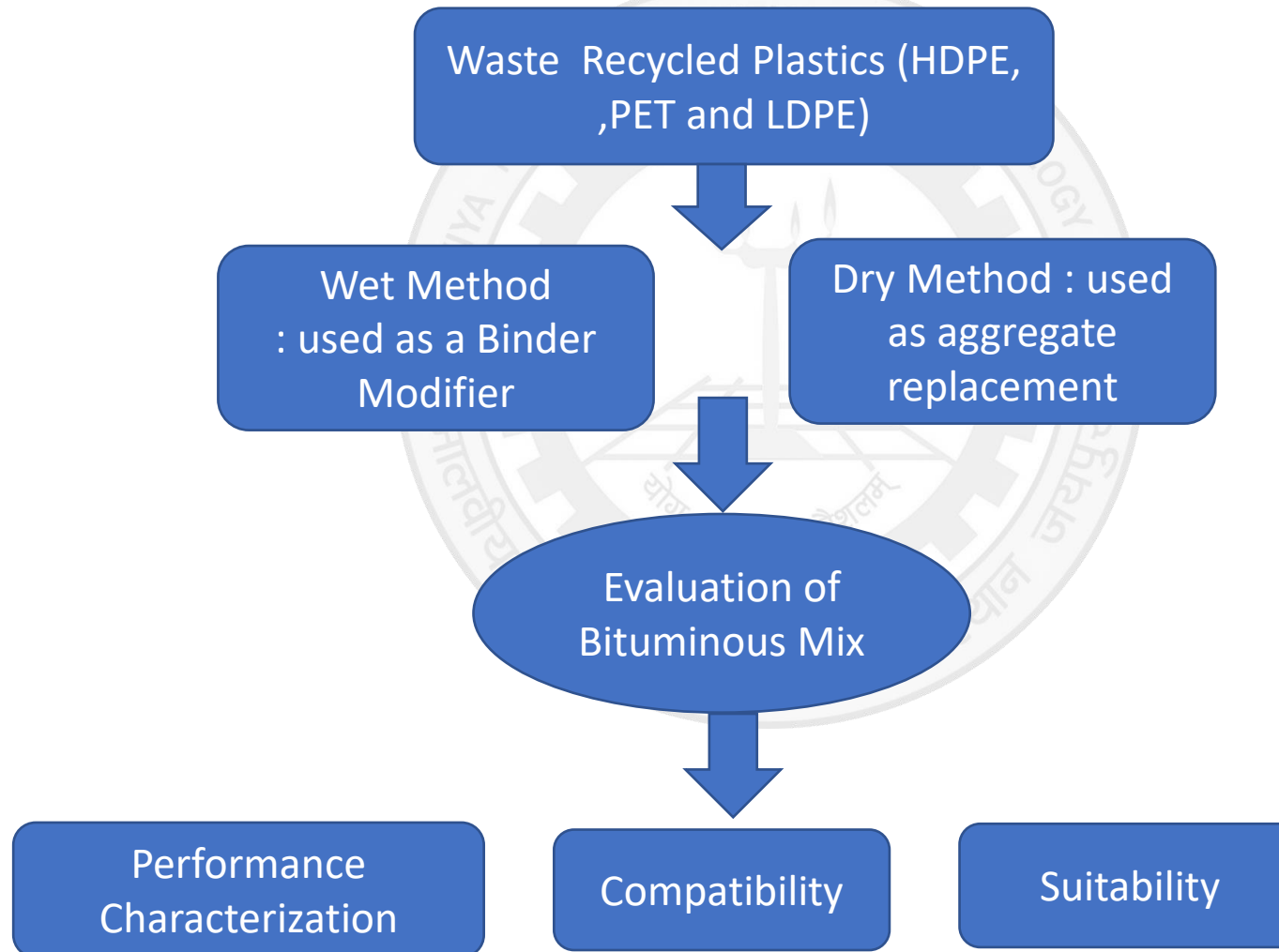
IRI & BI Graph



Findings & Proceedings for Part I

- Raveling is a Concern looking from I and II Cycle data.
- Cracking, Depression, Potholes, Edge Breaking are at better place In Plastic Roads than conventional roads.
- Condition ratings shows Plastic roads have slightly better ratings although almost all the roads have fair to good condition.
- Deflection data also suggests better findings for plastic Roads.
- Plastic Content on Constructed Plastic Roads is point of Concern.

Research Methodology :Comparison of Wet & Dry Mix Technology



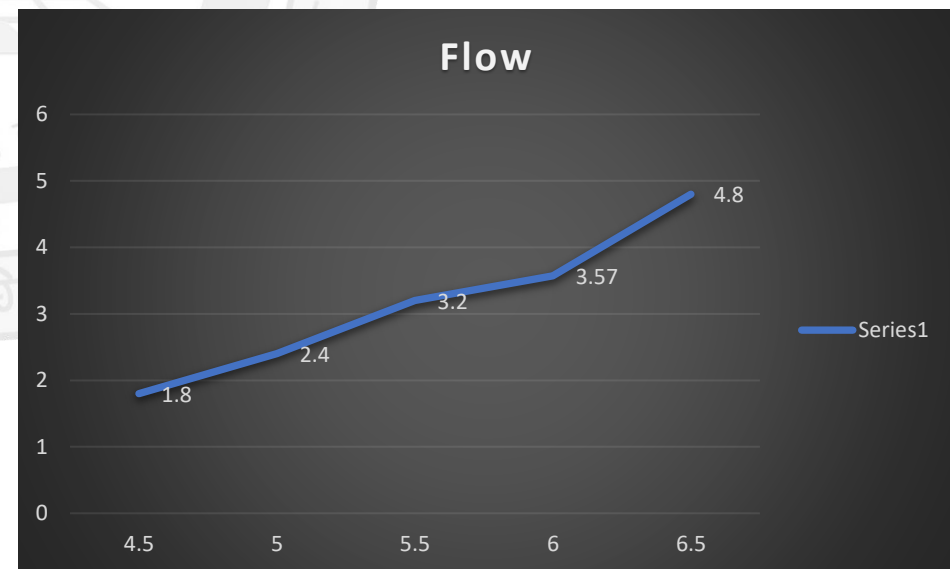
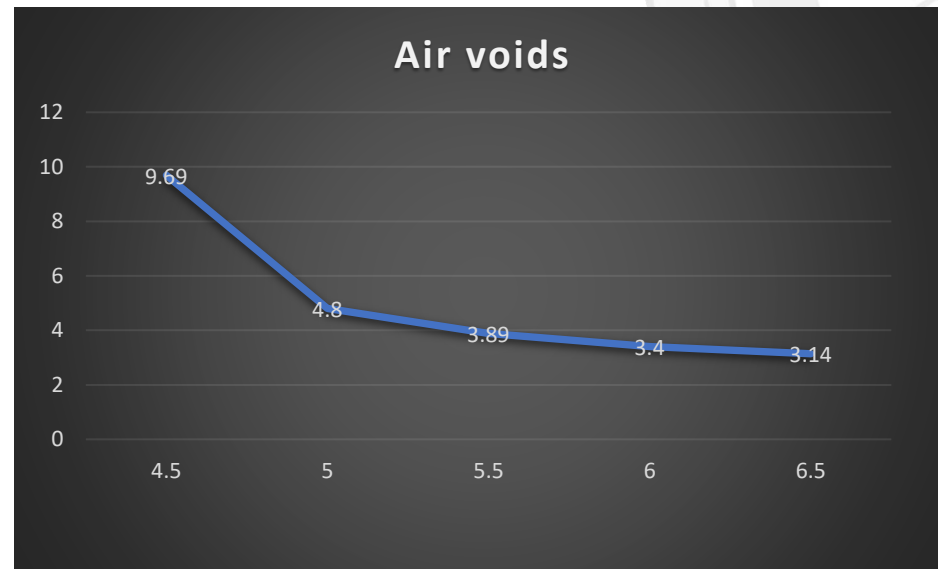
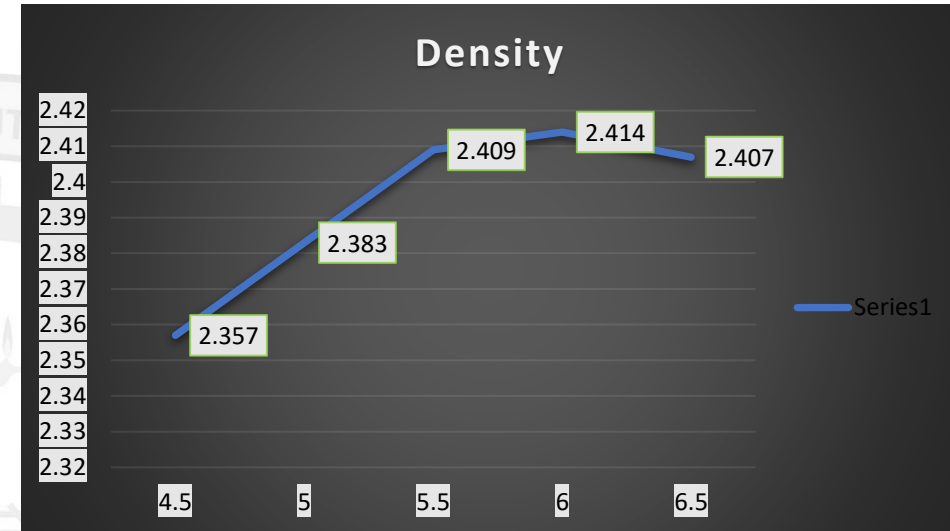
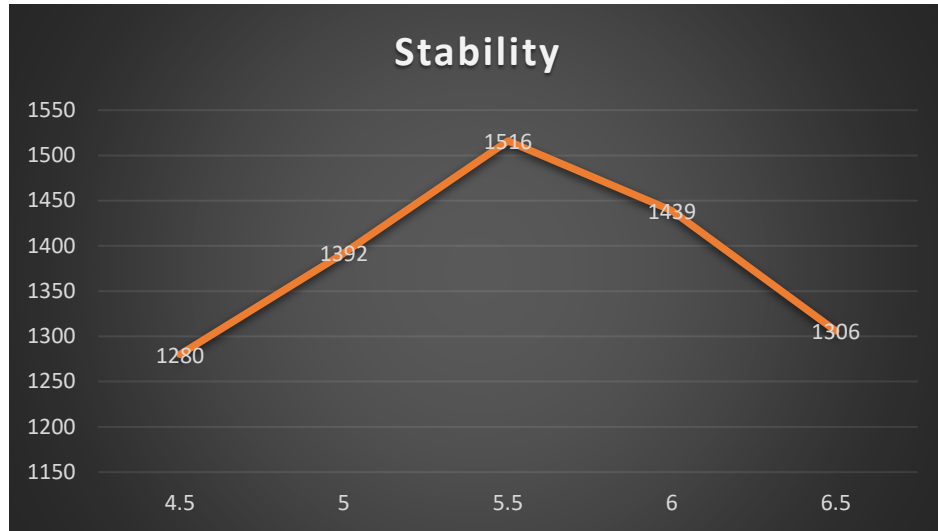
Aggregate Testing Results

S.No.	Test	Method	Value	Remarks
1.	Specific Gravity (20mm)	IS:2386 (part 3)	2.763	Satisfactory
	10mm		2.732	
	6mm		2.689	
	SD		2.65	
2.	Impact value	IS:2386 (part 4)	17.02 %	Satisfactory
3.	FI+ EI value	IS:2386 (part 1)	25.9	Satisfactory
4.	Water Absorption (CA)	IS:2386 (part 3)	0.37	Satisfactory
	FA		0.73	
5.	Stripping Value	IS:6241	>95%	Satisfactory

Natural Bitumen Results (VG40)

S.No.	Test	Method	Value	Remarks
1.	Specific Gravity	IS:1202	1.025	Satisfactory
2.	Penetration	IS :1203	40	Satisfactory
3.	Softening Point	IS: 1205	54.7*C	Satisfactory
4.	Viscosity	IS : 1206		Satisfactory
A.	Kinematic		515	
A.	Absolute		4135.5 Poises	
5.	Ductility	IS :1208	83.3 cm	Satisfactory

Marshall Design for BC-II



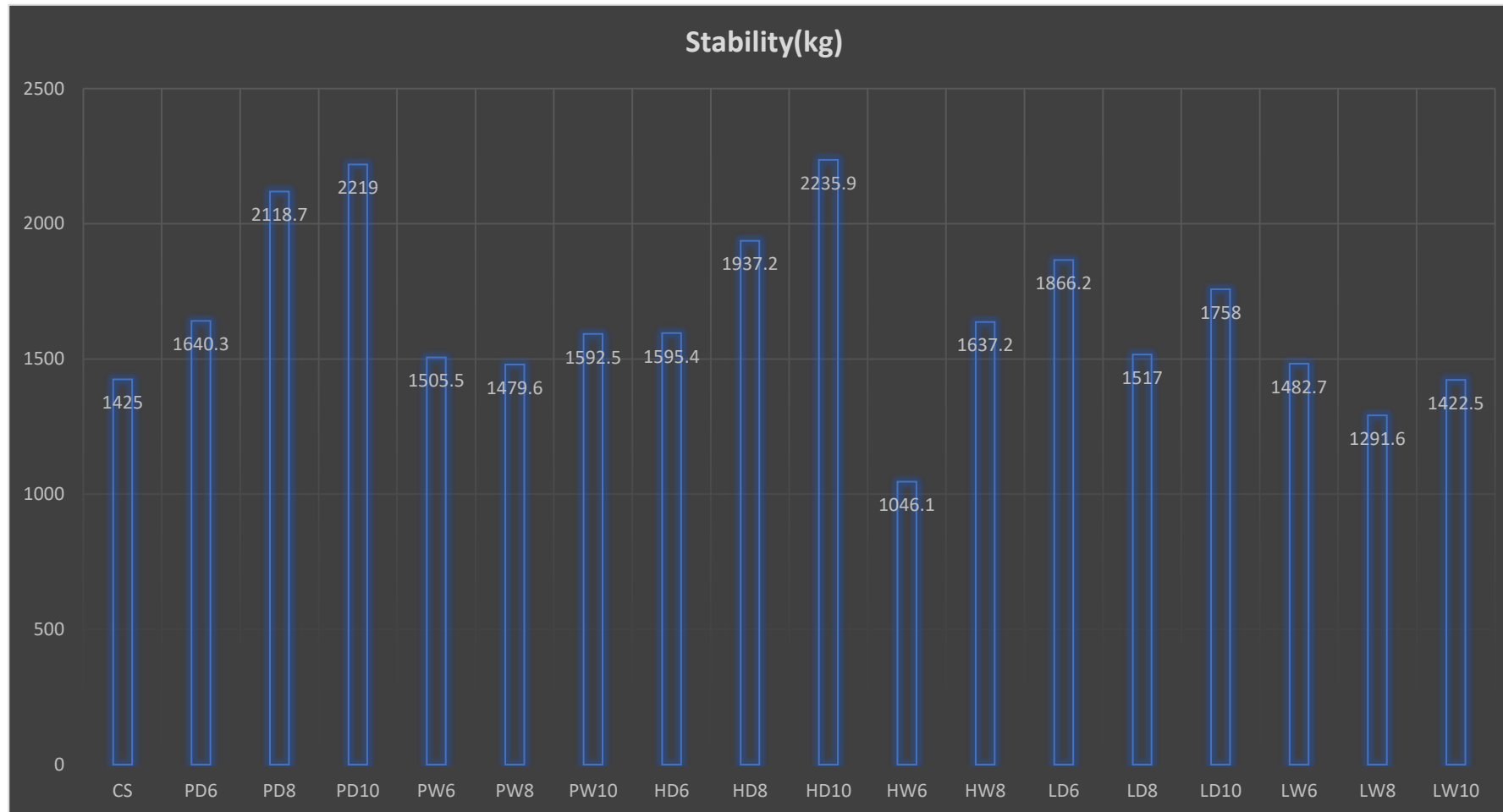
Dry Mix Design

Sample	Bulk density (gm/cc)	Air voids	VMA (%)	VFB	Stabilty (kg)	flow value(mm)
CS	2.471	3.59	13.05	72.49	1425	3.42
PD6	2.49	3.19	12.4	74.27	1640.3	2
PD8	2.488	3.27	12.47	73.78	2118.7	2.3
PD10	2.494	3.03	12.26	75.29	2219	3.3
HD6	2.514	2.26	11.56	80.45	1595.4	3.8
HD8	2.503	2.68	11.94	77.55	1937.2	3.2
HD10	2.504	2.64	11.91	77.83	2235.9	2.6
LD6	2.517	2.14	11.45	81.31	1866.2	2.3
LD8	2.498	2.88	12.12	76.24	1517	3.2
LD10	2.504	2.64	11.91	77.83	1758	3.5

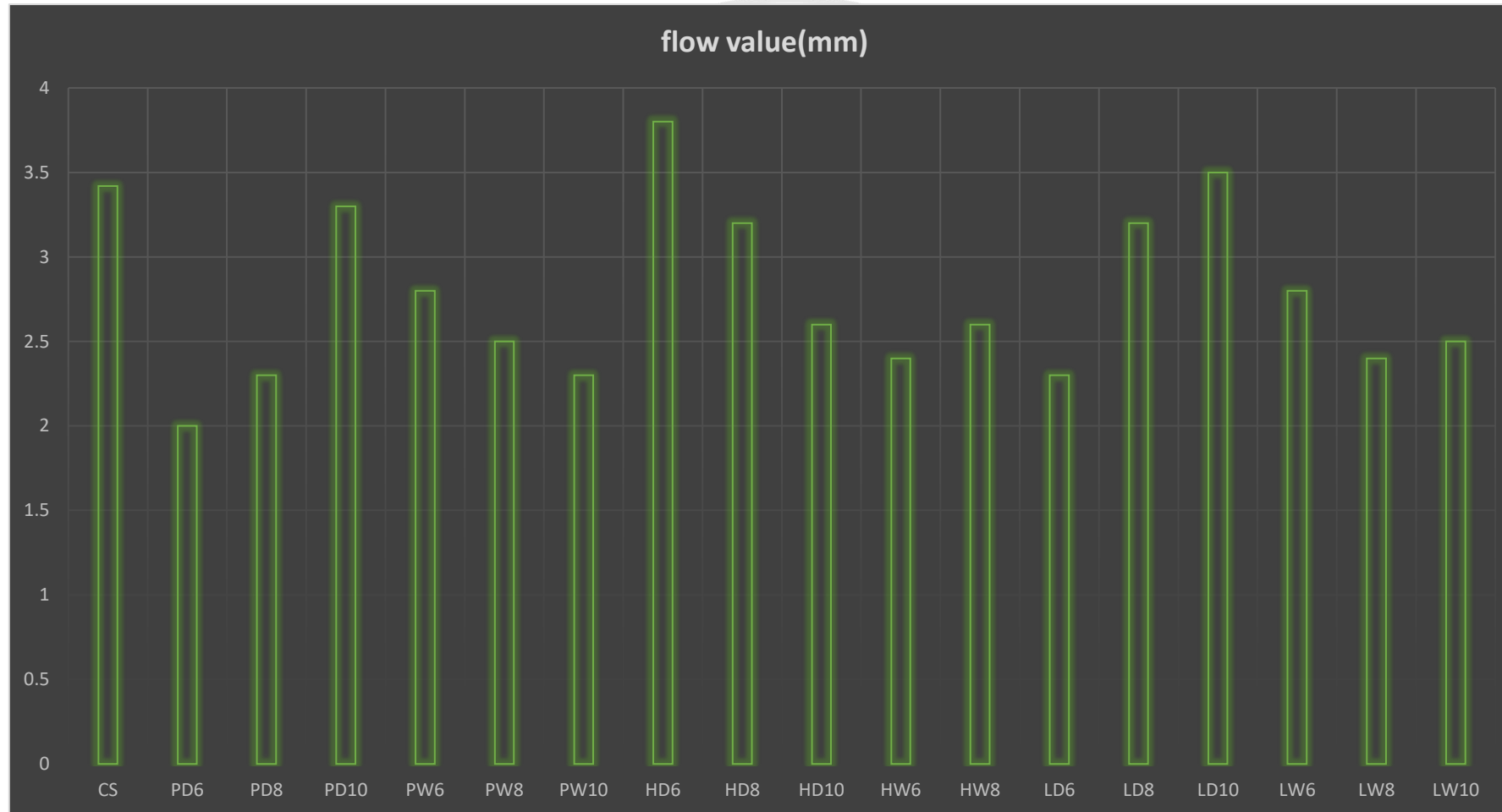
Wet Mix design

PW6	2.499	2.84	12.08	76.49	1505.5	2.8
PW8	2.486	3.34	12.54	73.37	1479.6	2.5
PW10	2.472	3.89	13.03	70.15	1592.5	2.3
LW6	2.487	3.3	12.51	73.62	1482.7	2.8
LW8	2.485	3.38	12.58	73.13	1291.6	2.4
LW10	2.478	3.65	12.82	71.53	1422.5	2.5
HW6	2.386	7.23	16.06	54.98	1046.1	2.4
HW8	2.438	5.21	14.23	63.39	1637.2	2.6

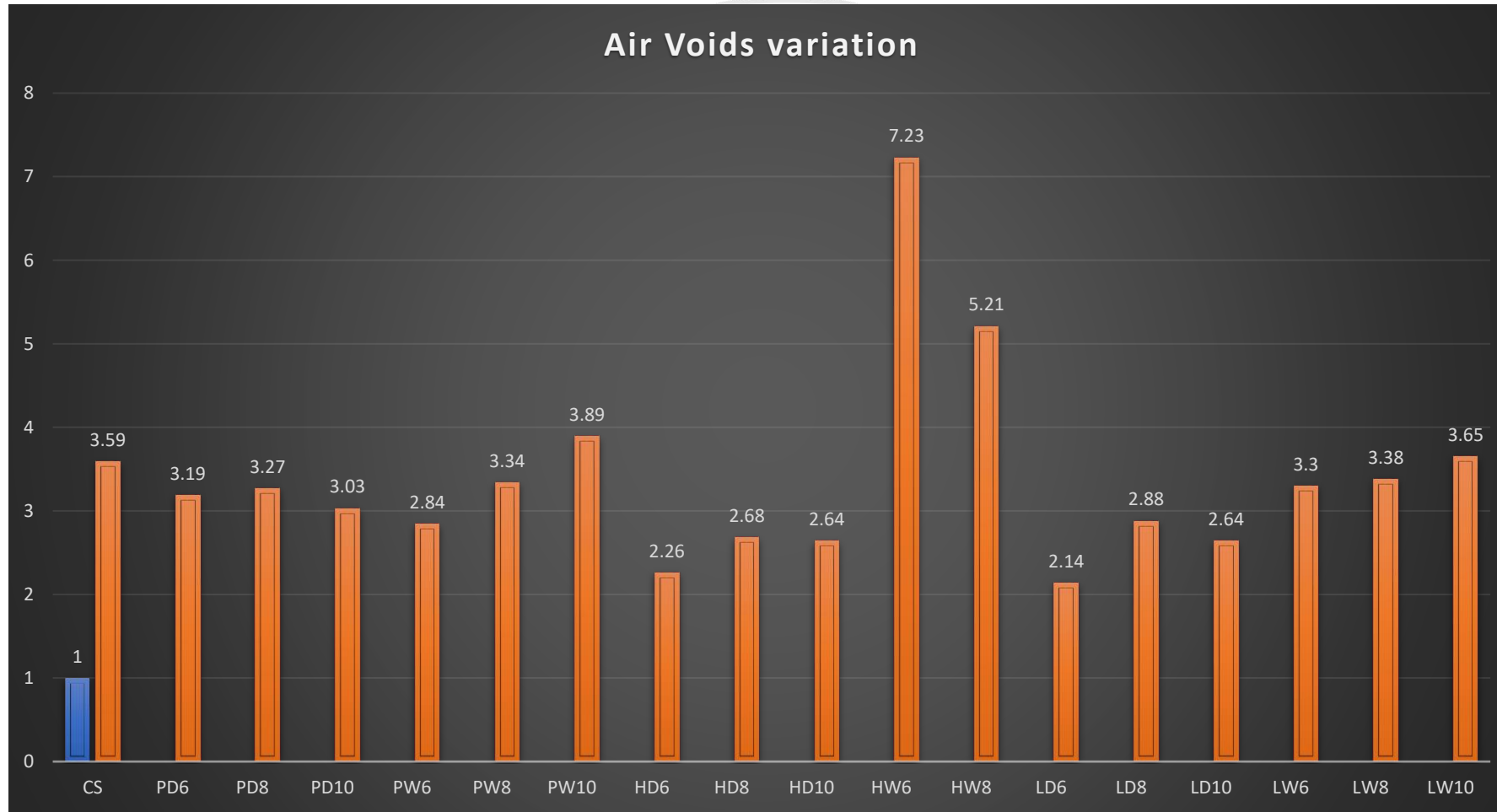
Stability Curve



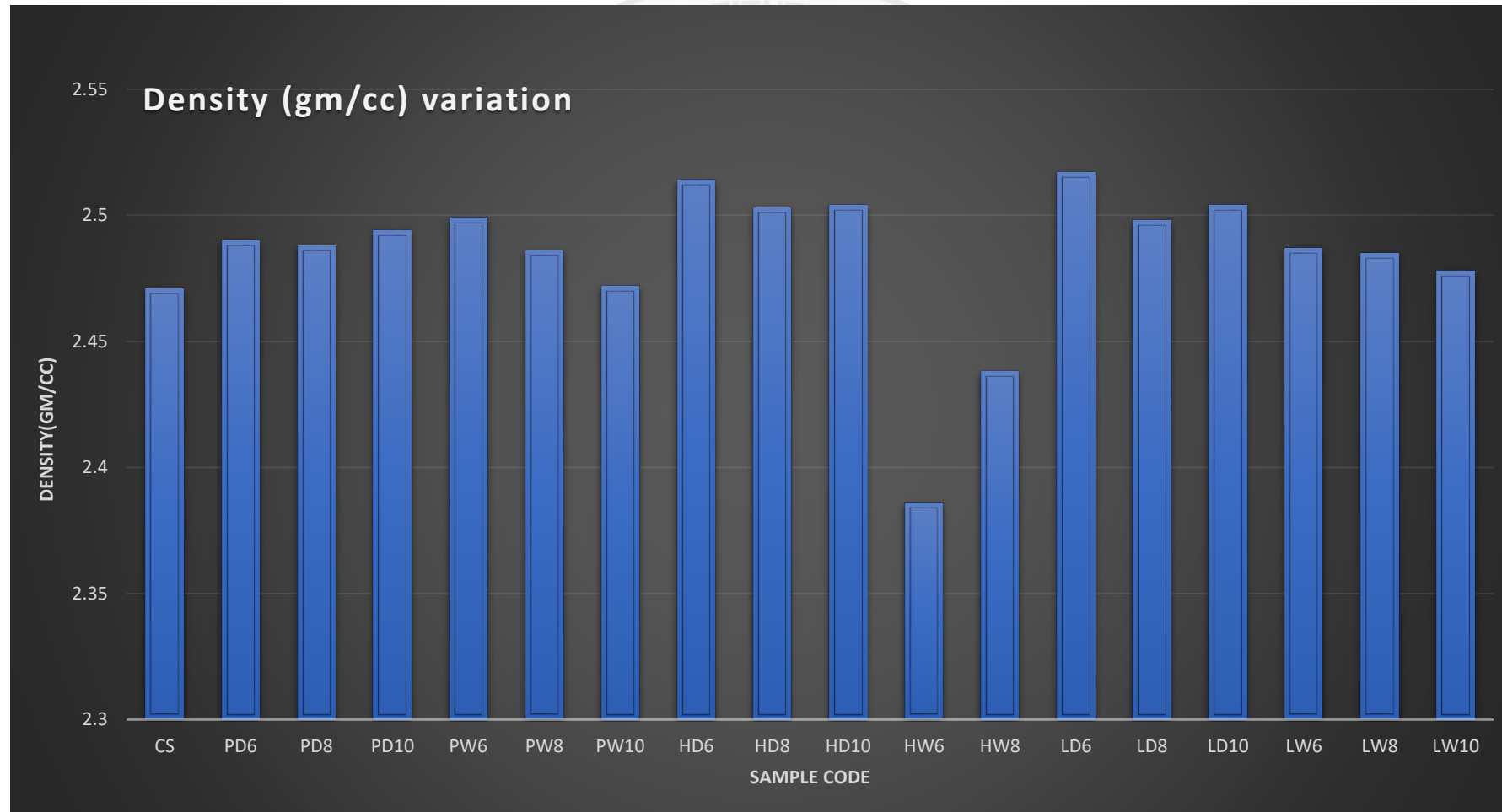
Flow Value Variation (mm)



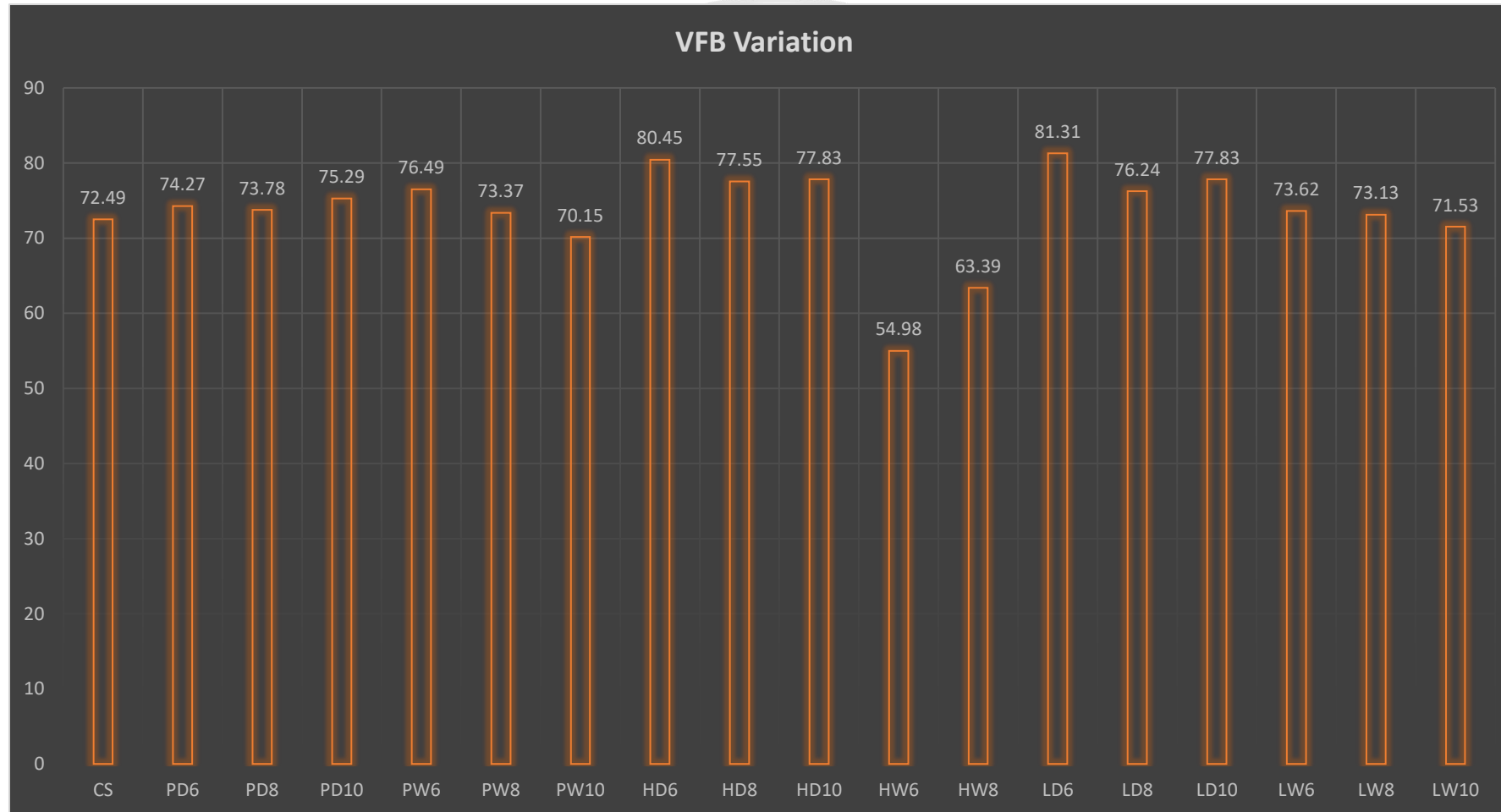
Air Voids Variation



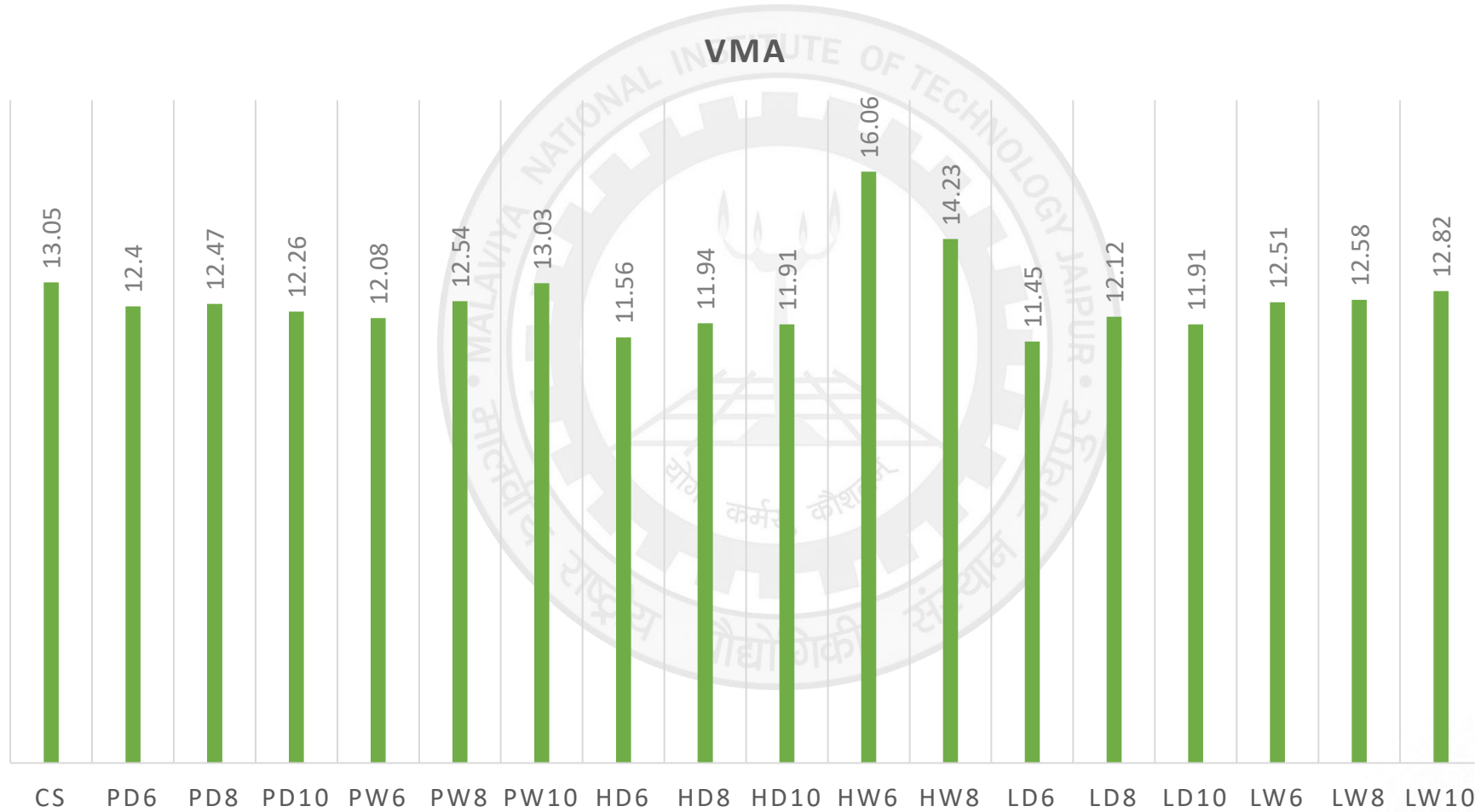
Density Variation



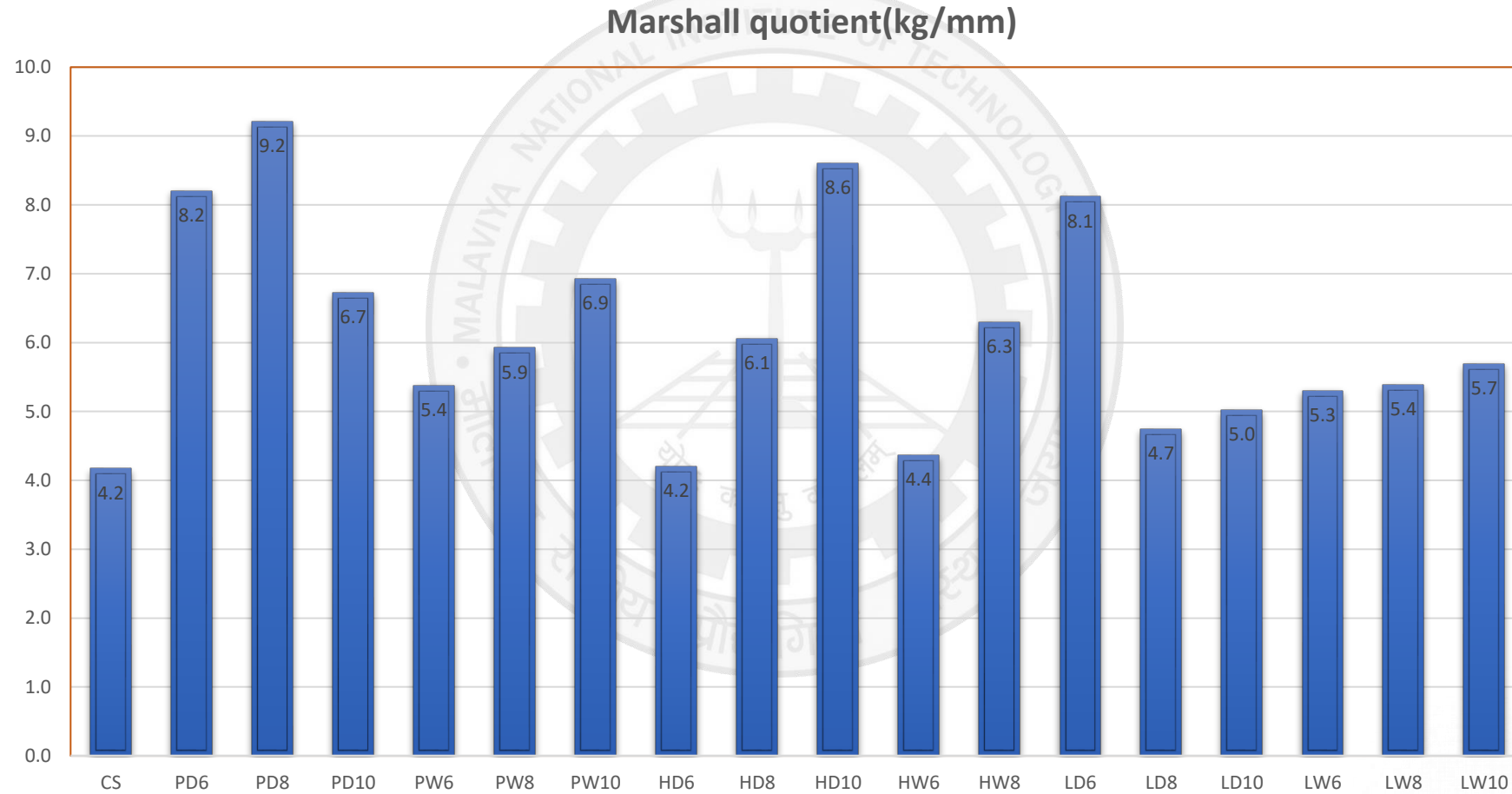
VFB Variation



VMA Variation

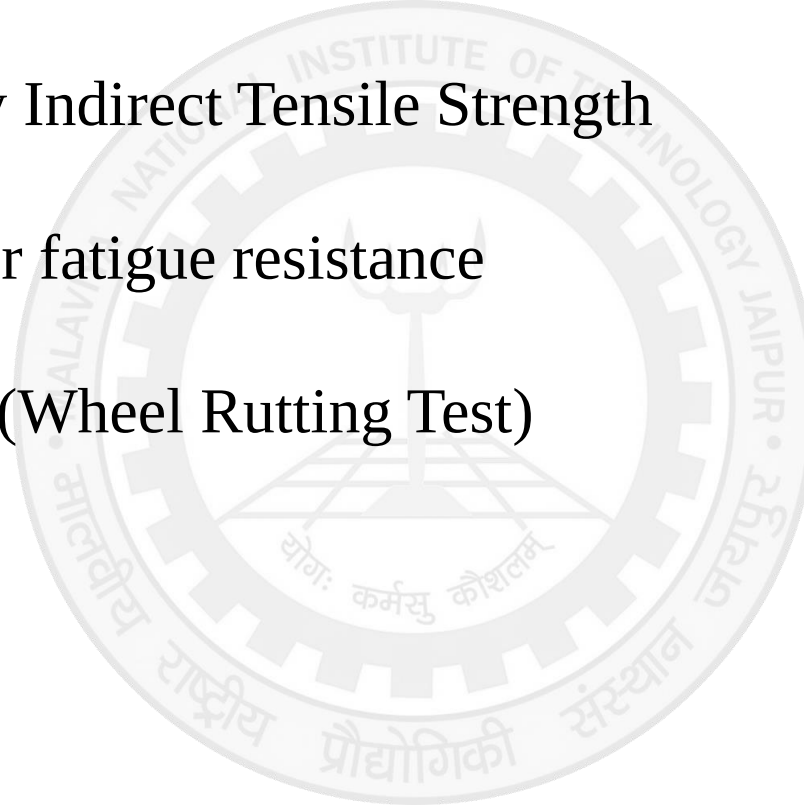


Marshall Quotient Variation



Performance Evaluation of BC Mixes by :

- A. Water Sensitivity by Indirect Tensile Strength
- B. Stiffness Modulus for fatigue resistance
- C. Rutting Resistance (Wheel Rutting Test)



Performance Evaluation (TSR)

The ITS test is used to evaluate the tensile strength and moisture susceptibility of asphalt mixtures.

It helps in understanding the potential for cracking and other failure modes in pavement materials.

The ratio of ITS in wet and Dry condition is Tensile Strength Ratio(TSR)

- TSR provides an indication of how well the material retains its strength when exposed to adverse conditions, such as moisture.
- High TSR values suggest good resistance to moisture-induced damage, whereas low TSR values indicate potential problems with moisture sensitivity, leading to issues like stripping and raveling.

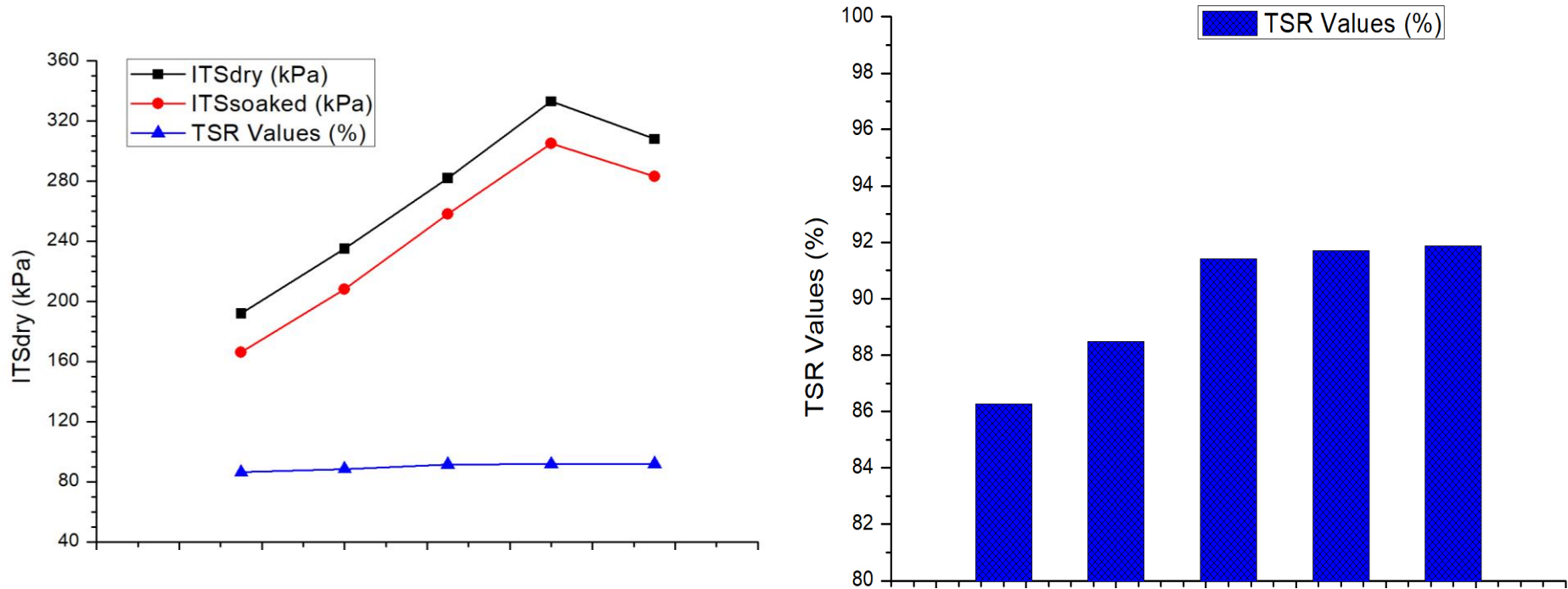
Indirect Tensile Strength Test



As per ASTM D 6931 (2012), a loading rate of 51 mm/min is used to test a specimen of diameter 101 mm on which load is applied through two vertically placed 13 mm wide loading strips.

ITS test Head and broken samples

Variation of ITS_{DRY} and ITS_{WET} along with respective TSR values for different mixes



Sample NO	CS	PD8	LD6	HD6	LW10
ITS_{DRY} (kPa)	192	235	282	333	308
ITS_{WET} (kPa)	166	208	258	305	283
TSR (%)	86.26	88.49	90.41	91.71	91.88

Analysis

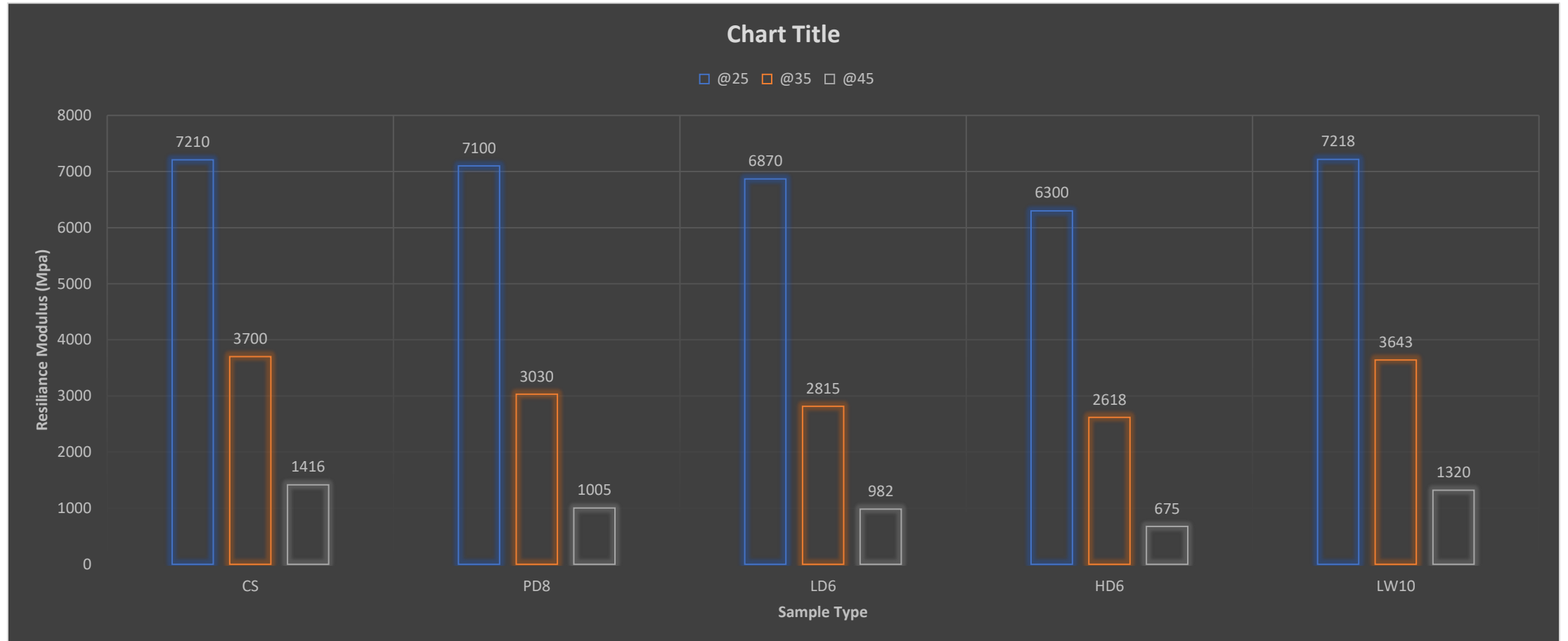
Resilient Modulus



Test Parameters	Value
Loading Pulse Width (ms)	100
Pulse repetition period (ms)	1000
Conditioning Pulse Count	50
Target Temperature (°C)	25, 35 , 45
Peak Loading Force (N)	10 % of ITS
Estimated Poisson's Ratio	0.35

UTM fitted with environmental chamber for resilient modulus test

Analysis : Modulus of Resilience



Analysis: Dynamic Creep

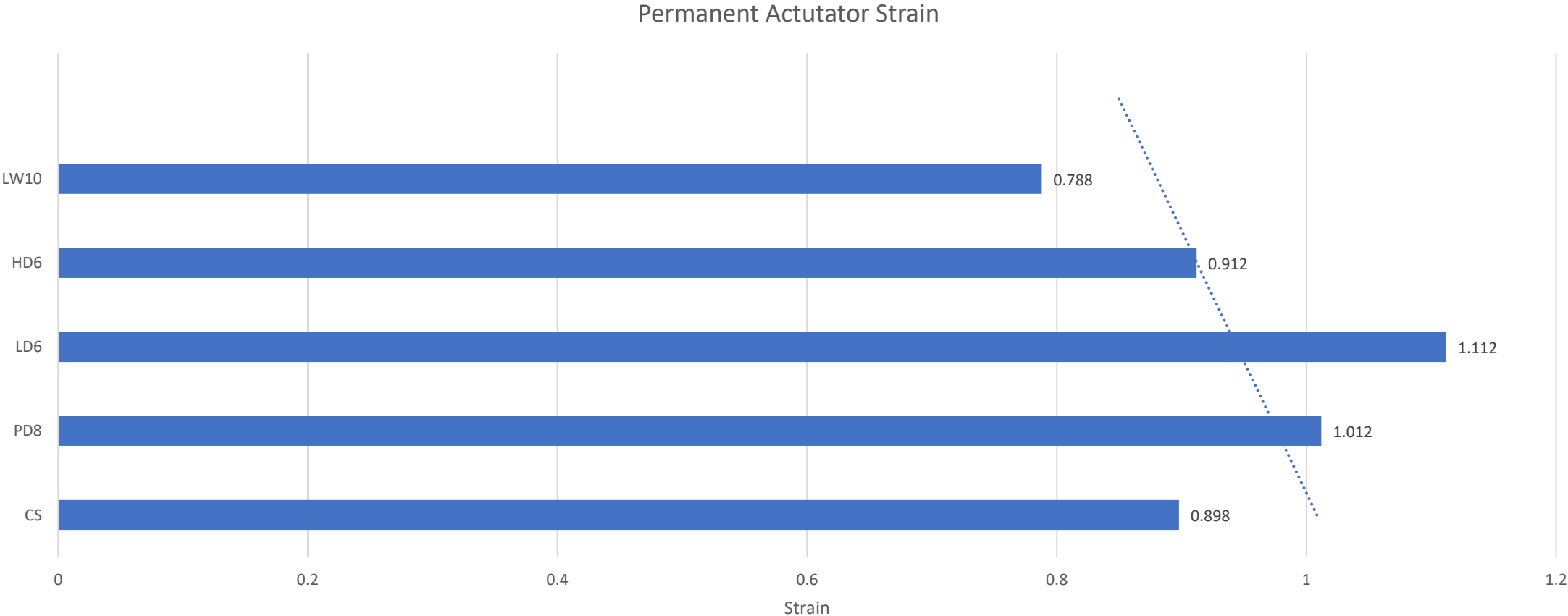
Dynamic Creep



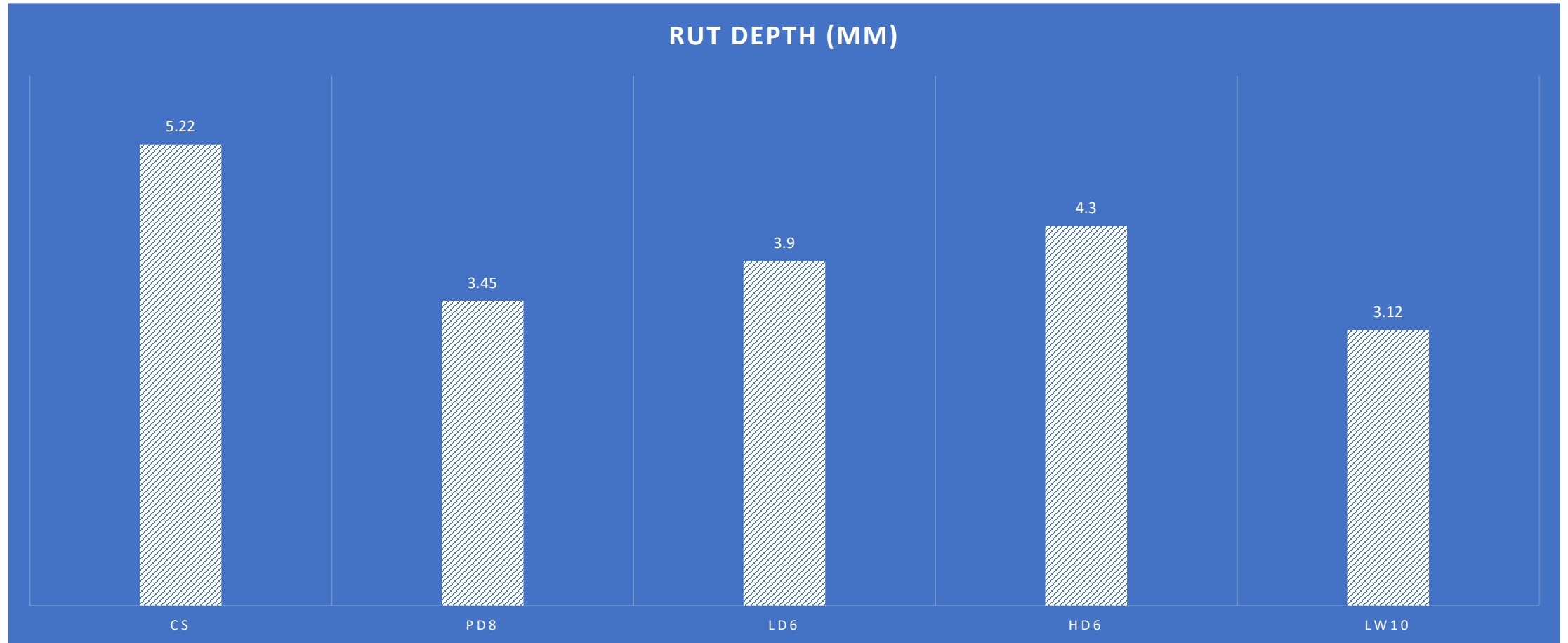
Test Parameters	Value
Loading function	Random
Cyclic stress (kPa)	200
Seating stress (kPa)	30
Load cycle width (ms)	1000
Load cycle repeat time (ms)	2000
Preload stress (kPa)	50
Preload time (s)	600

UTM for Dynamic Creep Test

Analysis : Dynamic Creep



Analysis : Wheel Rutting Test



Findings and Proceedings for Part II

- Performance Evaluation tests show better results for waste plastic mix samples in comparison of conventional samples.
- At mixing stage it was a problem to mix plastic in wet way as it require higher rate of mixing. PET ,LDPE and HDPE have a Easier approach in Dry mixing than Wet method.
- In the wet mix method increase of plastic percentage can be accumulated without any harmful impacts as well as size and shape related exercise can be avoided.
- Lab testing of samples suggests that Wet mixing is a better way For Polyethylene ie HDPE , LDPE although field analysis is a key factor for decision.\
- Further Microscopic Analysis by SEM and XRD techniques and Thermogravimetric Analysis (TGA) is also needed .

Publications

Published

1. DOI: [10.1088/1757-899X/872/1/012129](https://doi.org/10.1088/1757-899X/872/1/012129)

Second International Conference on Materials Science and Manufacturing Technology (ICMSMT 2020){Scopus Indexed}

2. DOI: [12.1088/1178SP/234/1/012142](https://doi.org/10.1088/1178SP/234/1/012142)

International Conference on New Materials & Technology) {Scopus Indexed}

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THANK YOU

